Chapter 1

Introduction

Chapter Outline

1.1 GLOBAL TRENDS IN ENERGY CONSUMPTION

In this section, historical energy consumption in the world and select countries as well as regions is first discussed based on reported data for primary energy supply and final energy consumption. Specific trends in current and future energy consumption and energy mix of the building sector are also outlined using available data and analyses. In addition, the economic output per unit energy consumed is evaluated to assess the macroeconomic energy efficiency of select countries and regions in the world.

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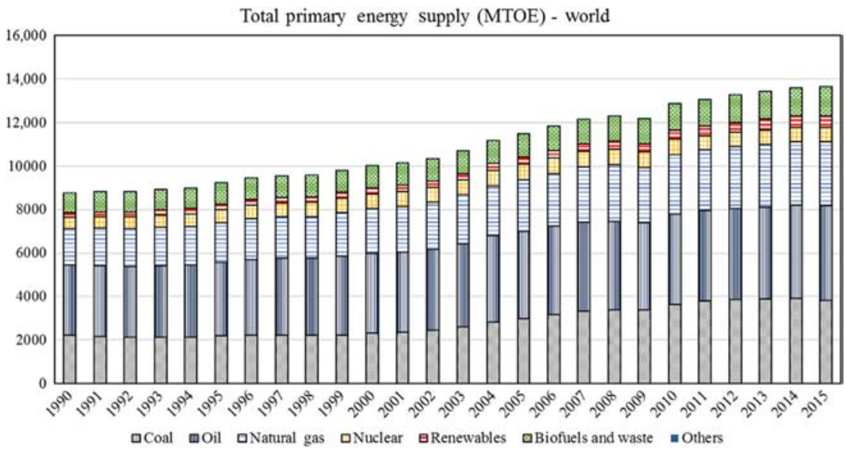

FIGURE 1.1 Annual variation of total primary energy supply (TPES) for the world from 1990 to 2015. *(Source: IEA, 2017).*

1.1.1 Primary Energy Supply

From 1990 to 2015, the total primary energy supply (TPES) in the world increased by 60% while the energy mix remained almost unchanged as noted in Fig. 1.1. Indeed, fossil fuels provided over 80% of the primary energy supply of the world during the last two decades. The specific composition of the world primary energy mix for 2015 is detailed in Fig. 1.2 (IEA, 2017). Specifically, the world relied mainly on oil (32%), coal (28%), and natural gas (22%) to meet its energy needs. Biofuels and waste (10%), nuclear (5%), hydro (2%), and

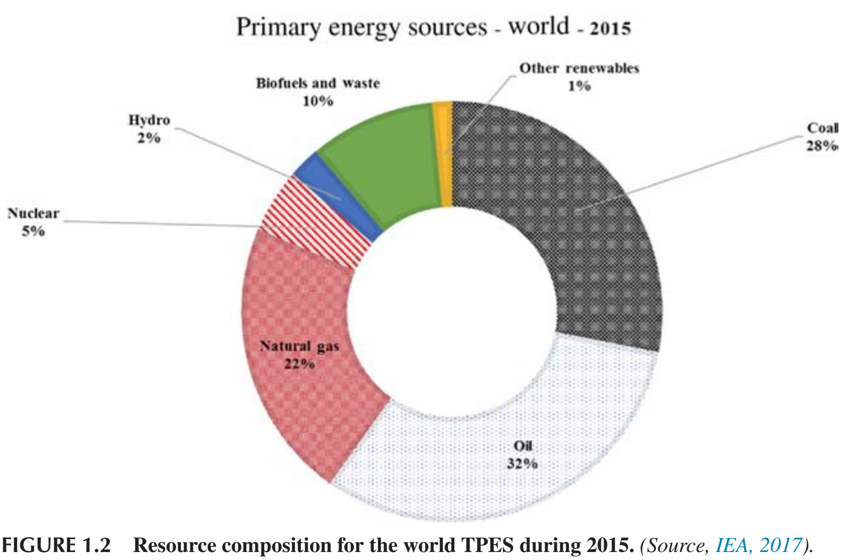

FIGURE 1.2 Resource composition for the world TPES during 2015. *(Source, IEA, 2017).*

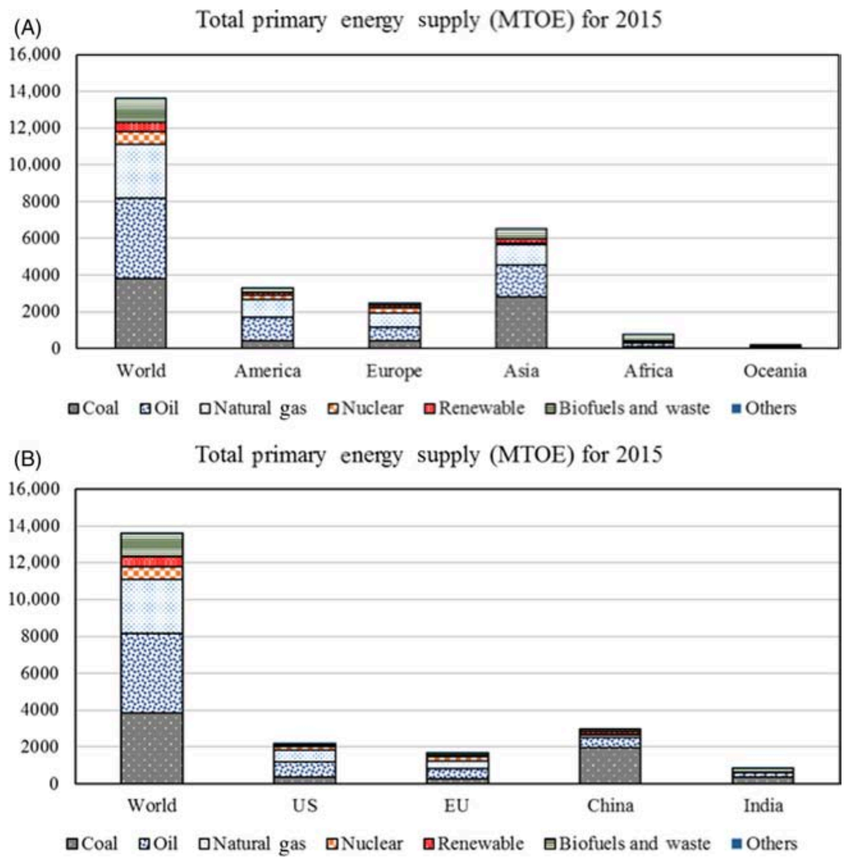

FIGURE 1.3 Primary energy resources for the world compared to those for (A) continents and (B) select economies, 2015. *(Source, IEA, 2017).*

renewable energy resources (1%) were also used to power the world economy during 2015. Fig. 1.3 compares the composition for TPES mix between continents as well as select countries and regions. China and Asia still rely heavily on coal to supply their primary energy needs while most of the other continents and major economies use predominantly oil and natural gas. During 2015, China and the US remained the top energy consumer countries with, respectively, 22% and 16% of the world TPES. Asia, mostly due to China and India, has the highest primary energy consumption among all the continents with over 47% of the world TPES.

As illustrated in Fig. 1.4, the annual world TPES increases at a slightly higher rate than that of the world population growth but significantly lower than the change rate of the world economic output measured by the gross domestic product (GDP). Indeed, the GDP has increased 60% from 1990 to 2015 indicat-

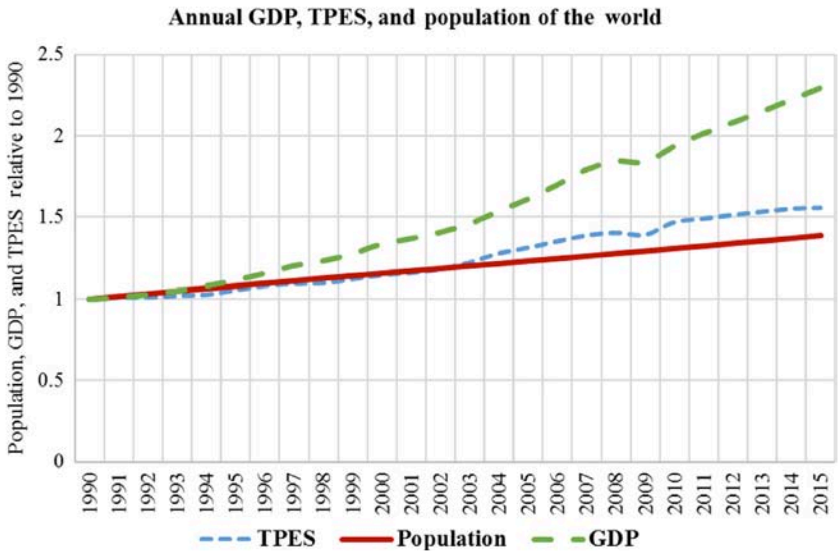

FIGURE 1.4 Annual change of gross domestic product (GDP), TPES, and population of the world indexed to 1990.

ing a decrease in energy intensity and thus an increase in energy efficiency and ultimately an increase in the energy productivity of the overall world economy as discussed in more detail in Section 1.1.4 and Chapter 5.

When normalized by the size of the population, the primary energy supply varies widely between countries as indicated in Fig. 1.5 for the world, US, EU, China, and India from 1990 to 2015 (IEA, 2017). Specifically, the world TPES

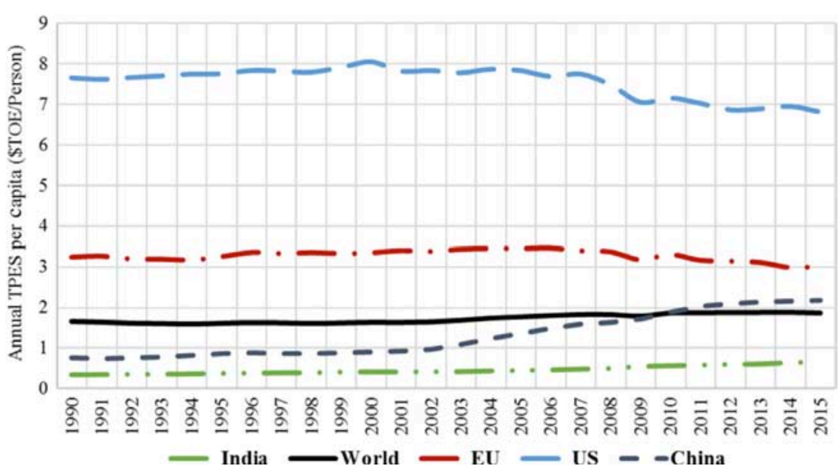

FIGURE 1.5 Annual variation of TPES per Capita for the world US, EU, US, China, and India.

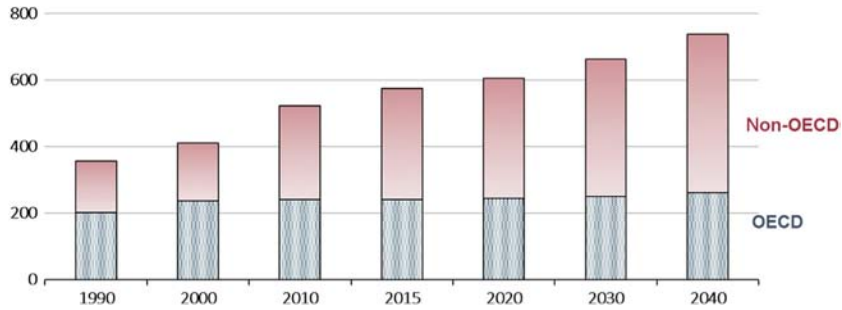

FIGURE 1.6 Predictions of annual TPES in quadrillion of Btu for the world including Organization of Economic Cooperation and Development (OECD) and non-OECD countries. *(Source, EIA, 2017).*

per capita has increased steadily and has reached an average of 2.0 TOE/person in 2015. The average energy use per capita has started lower than that of the world in both China and India but has increased steadily since the 2000s. In fact, TPES per capita in China has passed that of the world average in 2010 and reached 2.2 TOE/person in 2015. The energy use per person in India remains low at 0.75 TOE/person. The US had a significant energy use per person during the entire 1990–2015 period reaching a peak of 8.0 TOE/person in 2000 but declined since to settle at 7.0 TOE/person during 2015. A similar trend is observed for TPES per capita for the European Union (EU) but with a significant decrease since 2008 to be at 3.0 TOE/person for 2015.

Considering the current energy policies of major countries and economies, the US Energy Information Agency (EIA) has projected that future world TPES will continue to grow at a steady pace as illustrated in Fig. 1.6 showing past and future annual TPES for both OECD (i.e., countries part of Organization of Economic Cooperation and Development) and non-OECD countries (EIA, 2017). The EIA projections are based on the reference case with a 3.0% annual economic growth. Specifically, the EIA projects that the world energy consumption will reach 736 quadrillion Btu (i.e., 18,561 MTOE) by 2040, that is a 36% increase relative to the TPES recorded for 2015. As noted in Fig. 1.6, most of the energy consumption increase would be attributed to non-OECD countries including China and India. Using the projection case of high economic growth at 3.3%/year, an additional 40 quadrillion Btu (i.e., 1009 MTOE) is predicted to be consumed worldwide by 2040 relative to the reference case. For the low economic growth at 2.7%/year, the EIA expects that world TPES will be reduced by 29 quadrillion Btu (i.e., 731 MTOE) by 2040 compared to the reference case.

Based on the same EIA predictions as illustrated in Fig. 1.7, fossil fuels will remain the main resources utilized to meet the world's future energy needs. However, the overall share of fossil fuels in the world energy mix is predicted to decline. In particular, while petroleum and other liquids are predicted to remain the largest source of energy, their share in meeting world energy demand is

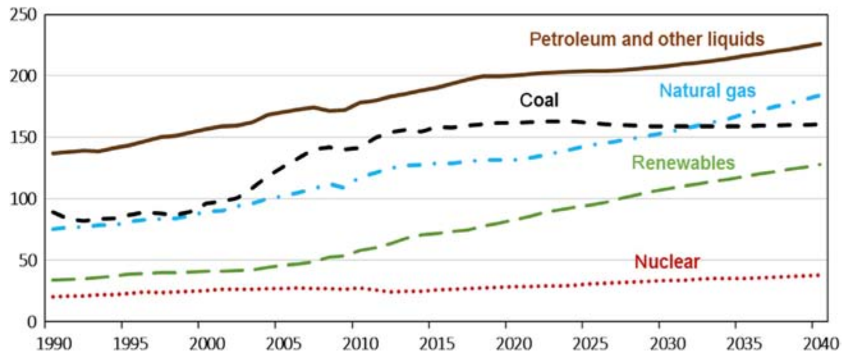

FIGURE 1.7 Predictions of resources of world primary energy consumption expressed in quadrillion of Btu countries. *(Source, EIA, 2017).*

expected to decrease from 33% in 2015 to 31% in 2040. The significant decline in the use of coal is projected to continue while the growth of using natural gas is predicted to increase at 1.4% per year. Renewables are projected to be the world's fastest growing energy resources in the next few decades. In particular, the share of renewables in the world's energy mix is projected to reach 17% by 2040.

1.1.2 Final Energy Consumption

Fig. 1.8 presents the annual variation and resource decomposition of total final energy consumption (TFC) by the world during the 1990–2015 period using reported IEA data (IEA, 2017). As noted for TPES, mostly fossil fuels and primarily oil products remain the main resources utilized to power the various

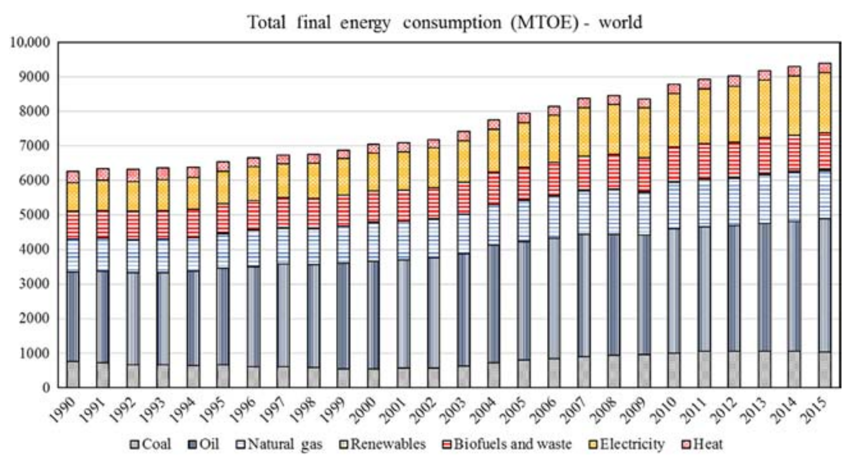

FIGURE 1.8 Annual total final energy consumption (TFC) and energy mix for the world between 1990 and 2015.

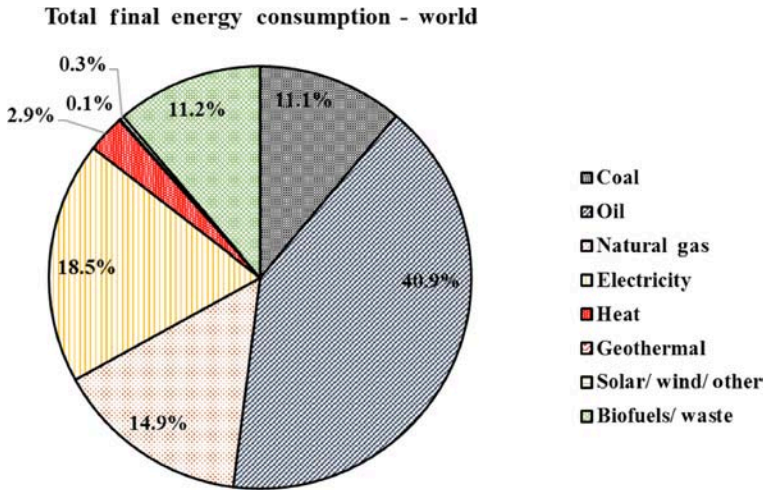

FIGURE 1.9 Energy mix of TFC for the world during 2015.

economic sectors of the world. During 2015, as illustrated in Fig. 1.9, the share of oil products was 40.9% of world TFC followed by electricity (18.5%), natural gas (14.9%), biofuels and waste (11.2%), and coal (11.1%). Renewables including geothermal, solar, and wind represented only 0.4% of the TFC in the world during 2015. Moreover, final energy resources are consumed evenly by three sectors: industry, transport, and buildings as illustrated in Fig. 1.10 through the 1990–2015 period. The share in world TFC of the building sector including residential buildings as well as commercial and public services has

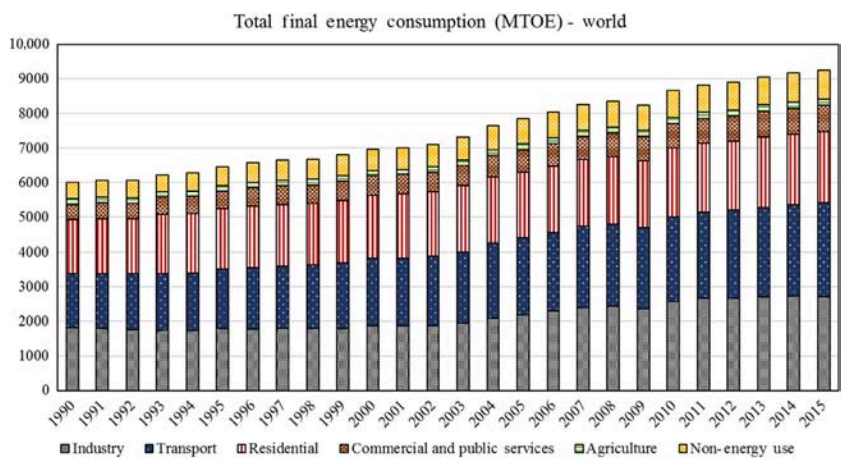

FIGURE 1.10 Annual world TFC variation by sector during 1990–2015.

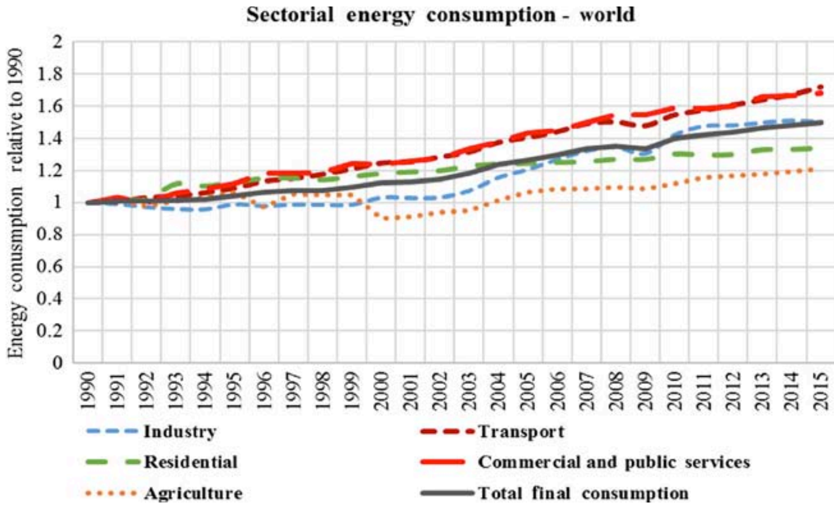

FIGURE 1.11 **Annual growth relative to 1990 of world TFC by sector during 1990–2015.**

reached 30% during 2015, the highest among all sectors. The growth in the TFC share of the building sector is mostly due to the increase in energy consumption associated with the commercial and public services as indicated in Fig. 1.11. While the overall TFC has increased by 50%, the energy consumed by commercial and service buildings has seen a growth rate of over 70% during the 1990–2015 period. Worldwide, the energy consumed by residential buildings has increased by 34%.

Fig. 1.12 illustrates the annual variation of the TFC as well as the ratio of TFC/TPES reported by IEA for the world during 1990–2015 period (IEA, 2017).

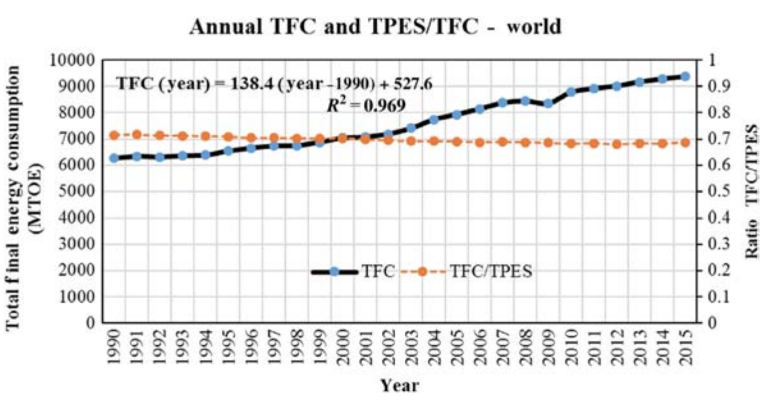

FIGURE 1.12 **Annual TFC and ratio of TPES/TFC for the world during 1990–2015.**

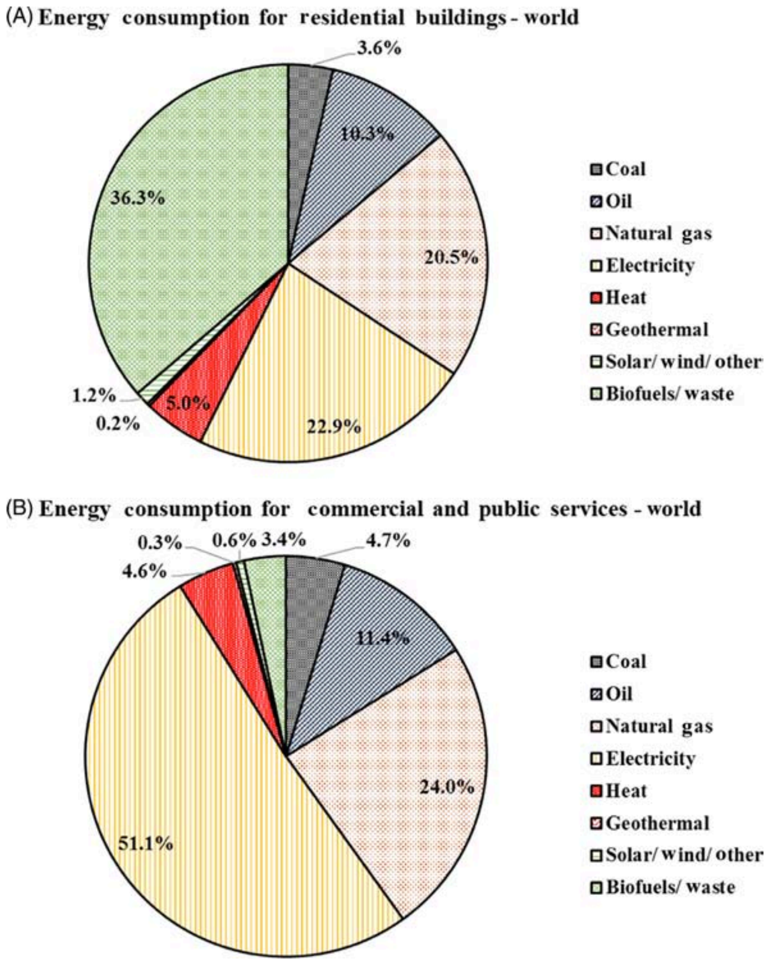

FIGURE 1.13 Share by resource for world's overall energy consumption for (A) residential buildings and (B) commercial and public services during 2015.

As noted in Fig. 1.13, a good linear correlation exists between TFC expressed in MTOE and the year after 1990 provided by the following regression:

$$TFC(year) = 138.4(year - 1990) + 5727.6 \, with \, R^2 = 0.969 \qquad (1.1)$$

The ratio of TPES and TFC remains almost constant during the 1990–2015 period slightly decreasing from 0.714 in 1990 to 0.687 in 2015. Example 1.1 illustrates the simple linear regression model of Eq. (1.1) to predict both TFC and TPES for the world in 2040. Using historical data, similar simplified models can be developed to predict future energy consumption for each region and country in the world (Krarti and Dubey, 2017; Krarti et al., 2017).

Example 1.1

Estimate the final energy consumption (TFC) and TPES for the world in 2040 using the simplified model of Eq. (1.1) and assuming that the TPES/TFC ratio is 0.68.

Solution

Using Eq. (1.1), TFC expressed in MTOE for year 2040 can be estimated:

$$TFC(2040) = 138.4(2040 - 1990) + 5727.6 = 12{,}648\,MTOE$$

Thus,

$$TPES(2040) = TFC(2040) \Big/ \left(\frac{TFC}{TPES}\right) = 12{,}648/(0.68) = 18{,}600\,MTOE$$

The model of Eq. (1.1) indicates that TPES is 737 quadrillion Btu, which is close to the EIA predictions based on the reference economic growth case.

1.1.3 Energy Consumption by the Building Sector

The building sector is responsible for the largest share of the world's TFC as indicated in Fig. 1.10. A wide range of resources is utilized to meet the energy demand for residential buildings and the commercial and public services. For the residential buildings, the world utilized during 2015 mostly biofuels and waste (36.3%), followed by electricity (22.9%), and then natural gas (22.5%) as indicated in Fig. 1.13A. For the commercial and public service facilities, the world consumed mostly electricity (51.1%), natural gas (24.0%), and oil (11.4%) as illustrated in Fig. 1.13B. However, the specific energy resources consumed by the building sector vary widely between countries and regions as shown in Fig. 1.14. For instance, biofuels and waste represent the dominant energy resource for residential buildings in India (73%) and to lesser extent in China (28%) but rarely used in the US (4%). On the other hand, natural gas is the energy resource of choice for households in the US (42%) and Europe (36%). For commercial buildings, electricity is the main energy resource for most countries and regions including the US (56%), EU (47%), China (34%), and India (35%).

Moreover, the contribution of the building sector in the national TFC varies significantly between countries and regions. Fig. 1.15 compares the sectorial decomposition of TFC for the world, US, EU, China, and India during 2015. The share of the overall TFC associated to the building sector including residential as well as commercial and public services is 34% for the US, 41% for the EU, 23% for China, and 41% for India during 2015. Within the same region of the world, the share of buildings in the national TFC can vary significantly as noted in Fig. 1.16 for some Middle East and North Africa (MENA) countries. Specifically, Fig. 1.16 shows the annual contribution of the building sector accounting for residential, commercial, and public buildings in national TFC

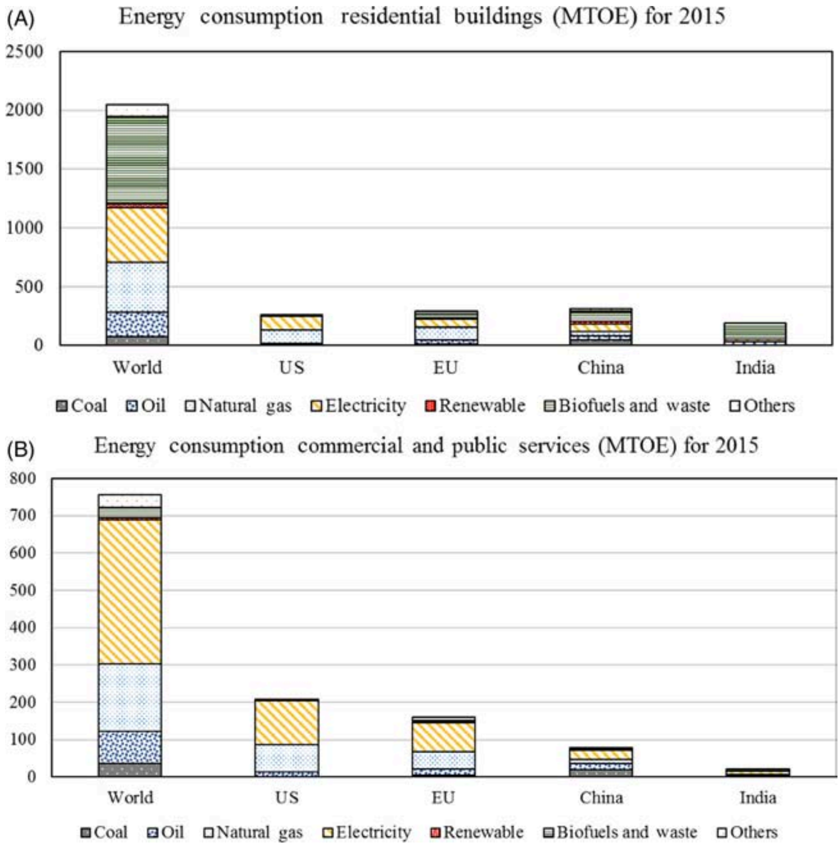

FIGURE 1.14 Resources for energy consumption for world, US, EU, China, and India attributed to (A) residential buildings and (B) commercial and public services during 2015.

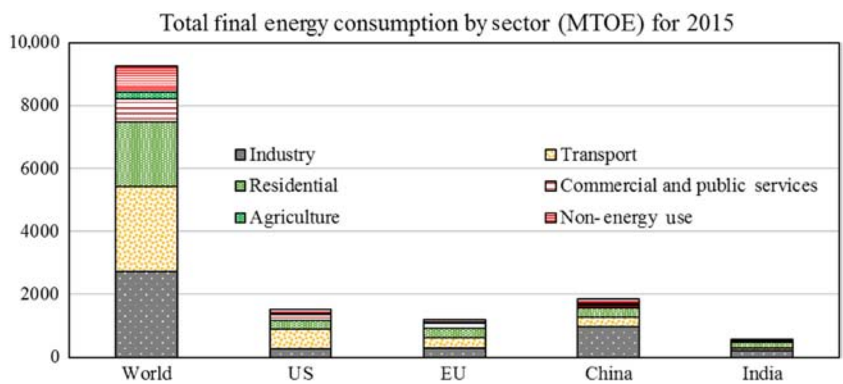

FIGURE 1.15 Sectorial TFC for the world, US, EU, China, and India during 2015.

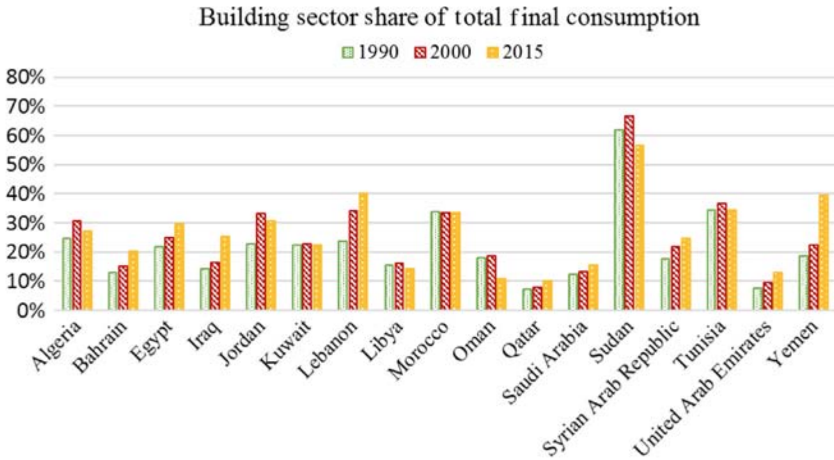

FIGURE 1.16 Share of the building sector in national TFC for select Middle East and North Africa (MENA) countries during 1990, 2000, and 2015.

during the years 1990, 2000, and 2015 for most MENA countries (IEA, 2017). Among the countries listed in Fig. 1.15, the TFC share of the building sector is highest for Sudan (57% in 2015) and is the lowest for Qatar (10% in 2015). For most MENA countries considered in Fig. 1.16, residential buildings consume more energy than commercial/public buildings as clearly indicated by the analysis shown in Fig. 1.17.

EIA projects that the energy consumption attributed to the world building sector, including residential and commercial facilities, will increase by 32%

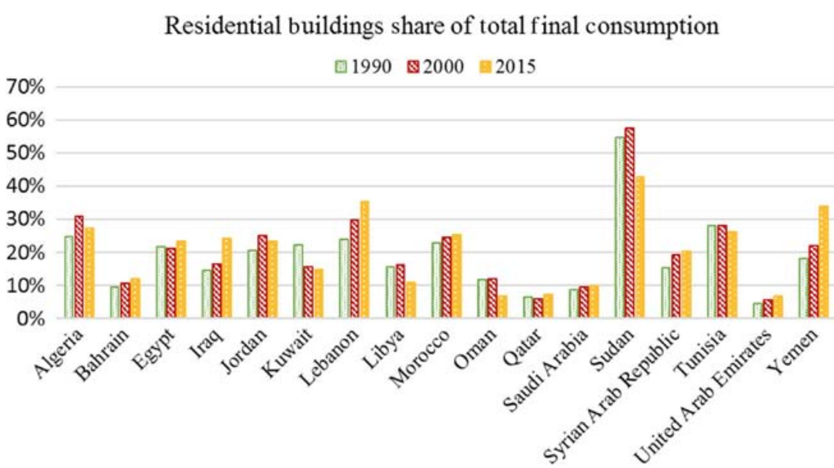

FIGURE 1.17 Share of the residential building sector in national TFC for select MENA countries during 1990, 2000, and 2015.

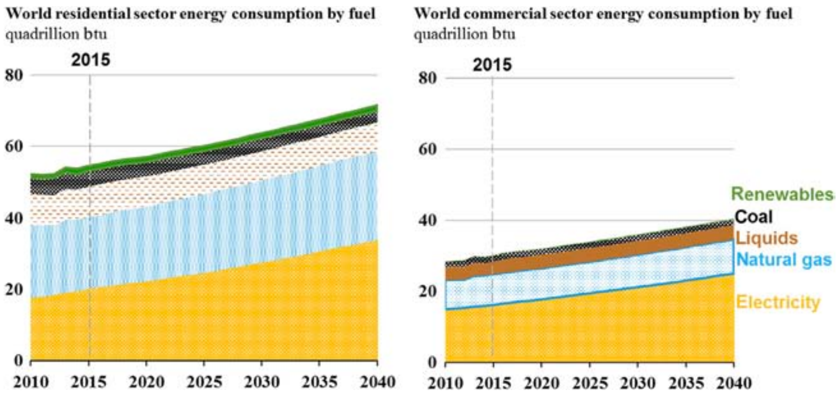

FIGURE 1.18 Projections for energy consumption of the world building sector. *(Source: EIA, 2017).*

between 2015 and 2040 as depicted in Fig. 1.18. It is expected that most of the energy use increase will occur in non-OECD countries especially in Asia including China and India that will experience increasing urbanization rates. Specifically, non-OECD Asia is expected to account for more than 50% of the world increase in the building sector. Between 2015 and 2040, electricity will be the fastest-growing source of energy used in both residential and commercial buildings with an annual growth rate of 2%. As illustrated in Fig. 1.19, the increase in electricity consumption will be driven by improving standards of living in non-OECD countries resulting in higher demand in appliances, lighting, cooling, and commercial services. The increase in energy consumption will offset the impacts of energy efficiency programs applied to buildings, appliances as well as heating and cooling equipment. Moreover, EIA projects that natural

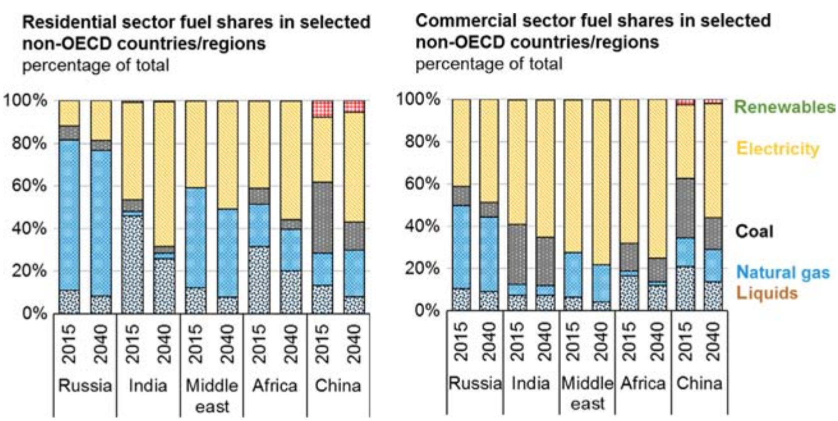

FIGURE 1.19 Distribution of energy resources used in the building sector by non-OECD countries and regions for 2015 and 2040. *(Source: EIA, 2017).*

consumption will increase by 20% during the 2015–2040 period mostly in non-OECD countries. The consumption of coal in the building sector is expected to decline further in the next decades.

1.1.4 Energy Productivity Trends

Defined as the inverse of energy intensity, that is, the economic output per unit of energy consumed, energy productivity has been used as a macroeconomic indicator to assess the balance between economic growth and domestic energy consumption (KAPSARC, 2015). Specifically, energy productivity is often utilized to determine the level of structural diversification between energy-intensive and nonenergy-intensive activities as well as to estimate the overall energy efficiency of an economy. However, as detailed in Chapter 5, energy productivity can be the basis for a framework analysis for energy programs to account for their overall benefits and value added. In particular, the energy productivity-based framework can incorporate the multiple benefits of energy efficiency including economic, environmental, and social. Indeed, an energy productivity framework can include traditional benefits (i.e., reduced energy consumption) as well as nontraditional value added of energy efficiency such as increased asset values, reduced greenhouse gas emissions, and creation of jobs as discussed in more detail in Chapter 5.

Fig. 1.20 presents the annual variation of the GDP per capita in the world, US, EU, China, and India during the 1990–2015 period. It is clear that the economic output per person in the world has steadily increased to reach $14,322/person during 2015. In terms of actual GDP per capita, average value for the world, China, and India have remained significantly less than similar indicators of $34,580/person and $51,592/person reported for the EU and the US,

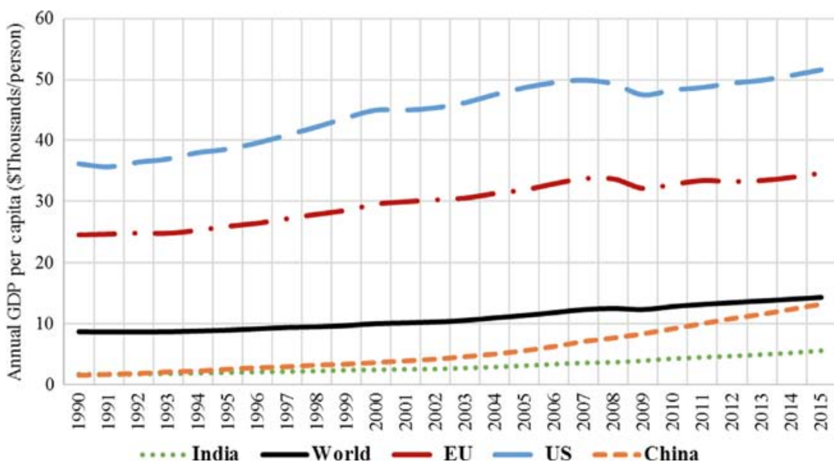

FIGURE 1.20 Annual GDP per capita for the world, US, EU, China, and India during 1990–2015.

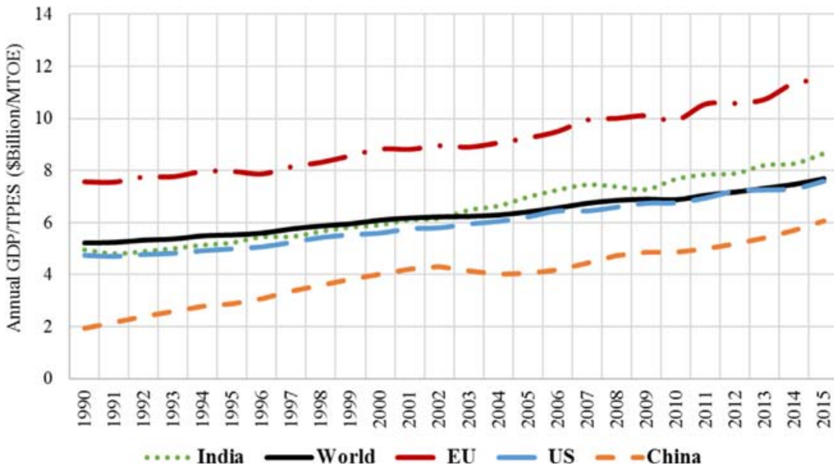

FIGURE 1.21 Annual energy productivity set as the ratio GDP/TPES, for the world, US, EU, China, and India during 1990–2015.

respectively. However, the growth rate of the per capita economic output for China, India, and the world has been higher than that for the EU and the US. In particular, the improvement of GDP per capita has increased by 789% for China, 223% for India, and 65% for the world well above the increase reported for the EU (41%) and the US (42%) from 1990 to 2015.

Because of the trends noted for GDP and TPES, the overall energy productivity, defined as the ratio of GDP and TPES, has been increasing in the world as well as the US, EU, China, and India as shown in Fig. 1.21. The energy productivity has steadily increased between 1990 and 2015 in the world as well as the US, EU, China, and India. Specifically, energy productivity for the world has increased by 47.7% from $5.2 billion/MTOE in 1990 to $7.7 billion/MTOE in 2015. During the same period (i.e., from 1990 to 2015), the growth of energy productivity differed between countries and is 51.7% for EU, 60.2% for US, 74.9% for India, and 214.5% for China due to the significant increase in its economic output during the same period.

For several countries, the main driver between the relationship between GDP and energy consumption is the contribution level of the industrial sector to the economy. Indeed, energy-intensive industries are often linked to the early stages of the development of an economy. As the income increases, the economy typically transitions to less intensive industries and sectors such as services. Thus, the more an economy relies on the industrial sector for its development, the more energy it consumes. However, the use of energy-efficient processes and systems can reduce the high-energy dependency to produce wealth. Fig. 1.22 illustrates the annual variation of GDP with total and sectoral final energy consumption for select countries including the US, UK, China, and India over the 1990–2013 period. A strong relationship exists between GDP and TFC

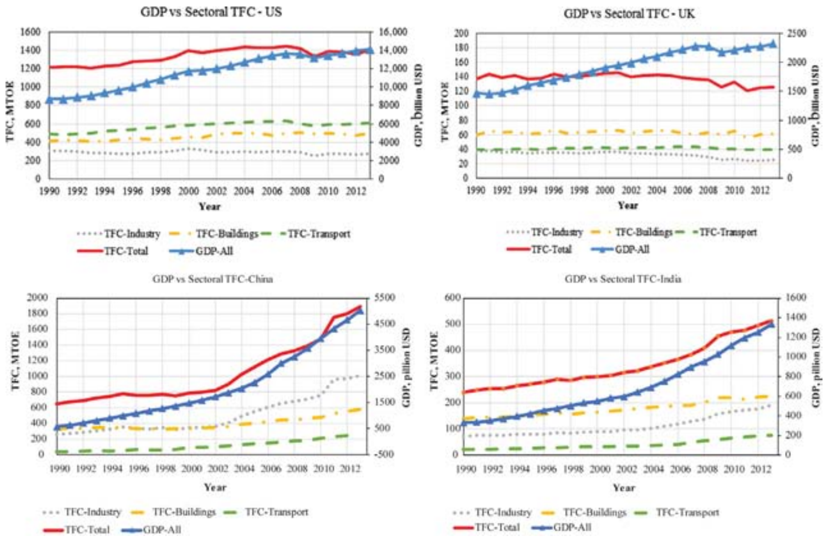

FIGURE 1.22 Annual variation of GDP and energy consumption for industry, transport, and building sectors for US, UK, China, and India.

for China and India, and to a lesser extent for the US. For the UK, however, TFC has decreased since 2000 while GDP continues to increase highlighted by a reduction in energy consumption for all sectors of its economy.

Fig. 1.23 provides the results of a regression analysis between GDP and TFC for the US, UK, China, and India during the period 1990–2013. As expected, good correlations exist between the GDP and TFC for China, India, and the US. In particular, the slope of the linear regression indicates that each unit of energy

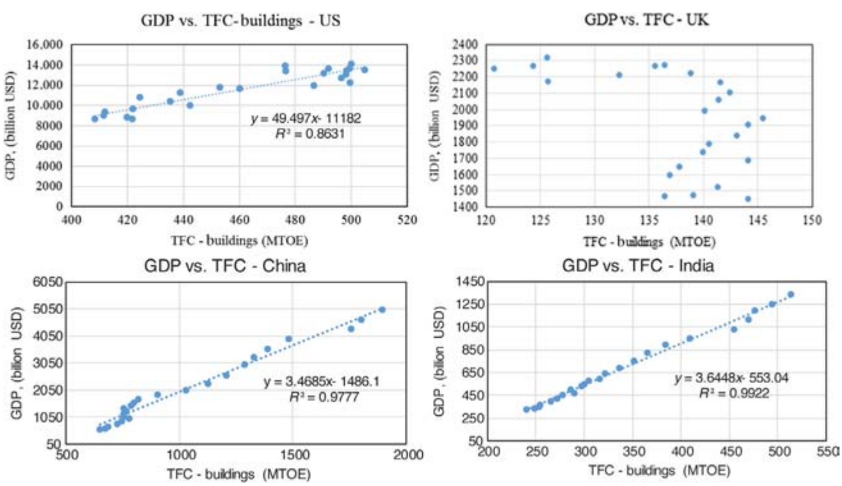

FIGURE 1.23 Regression between GDP and TFC for select countries.

consumed (in MTOE) has increased the GDP by $49.50 billion for the US, but only by $3.64 billion for India, and $3.46 billion for China. The relationship between GDP and TFC is weak for the UK. In fact, the UK follows the expected behavior referred to as the environmental Kuznets curve (EKC) hypothesis.

In econometrics, the EKC hypothesis has been considered by several studies to evaluate the relationship between economic growth and environmental pressure. In particular, Grossman and Krueger (1991) found that there is an inverted U-shaped relationship between income level and environment deterioration. This relation was termed the environmental Kuznets curve or EKC by Panayotou (1993) based on the original work on Kuznets curve indicating an inverted U-shaped correlation between income and inequality. In several studies, the total energy used by a country (expressed typically in terms of energy consumption per capita) is used as an indicator of environmental pressure while GDP (expressed in GDP per capita) is often used to measure the economic growth for the same country (Streimikiene and Kasperowicz, 2016). Some studies considered carbon emissions as an alternative indicator for environmental pressure (Kaika and Zervas, 2013a, 2013b).

According to the EKC hypothesis, the environmental pressure increases with the economic growth until a turning point is reached for a specific income per level. After this turning point, further economic growth leads to improved environmental quality. The results of the EKC analysis are often used to provide policymakers with a better understanding of the link between energy consumption and economic growth and ultimately assist them to formulate sustainable energy policies.

Using the annual energy consumption per capita, $x = E/P$, and the annual economic output expressed GDP per capita, $y = Y/P$, the EKC analysis consist of finding the correlation coefficients generally based on the model defined by Eq. (1.2):

$$e_{i,t} = \alpha_i + \gamma_t + \beta_1 y_{i,t} + \beta_2 y_{i,t}^2 + \beta_3 y_{i,t}^3 + \varepsilon_{i,t} \tag{1.2}$$

where the index i refers to a specific country for which the data and the analysis is carried out, the index t refers to the year when the data are collected, α_i is a constant coefficient specific to the country i, γ_i is a constant coefficient specific to the year t, β_1, β_2, and β_3 are coefficients correlating e to y as noted in Eq. (1.2), and $\varepsilon_{i,t}$ is the error term of the correlation.

Instead of e and y, some studies have used $Ln(e)$ and $Ln(y)$ to estimate the regression coefficients of Eq. (1.2) to avoid zero and negative indicators (Kaika and Zervas, 2013a). Fig. 1.24 illustrates the relation between the e and y as stipulated by the EKC hypothesis and modeled by the correlation of Eq. (1.2). At the macroeconomic scale, the energy consumption increases with the income level as the country is developing from a low-tech farming economy to an energy-intensive industrial economy reaching a turning point. With further development and transition to a less energy-intensive service economy, the energy consumption decreases as the income level continues to increase. For sustainable development, it is desired to increase the economic growth while reducing

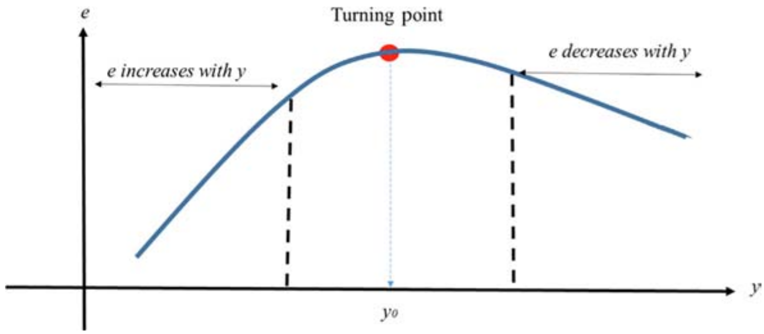

FIGURE 1.24 Typical relationship between energy consumption, *e*, and income level, *y*, following EKC model.

energy consumption as well as carbon emissions, and ultimately improving the environmental quality.

Considering the possible values of the regression coefficients, β_1, β_2, and β_3 in the model of Eq. (1.2), seven scenarios can be obtained but only one of them would satisfy the EKC hypothesis as listed here (Dinda, 2004):

- Scenario 1: if $\beta_1 = \beta_2 = \beta_3 = 0$, there is no relationship between e and y.
- Scenario 2: if $\beta_1 > 0$ and $\beta_2 = \beta_3 = 0$, an increasing linear relationship exists between e and y.
- Scenario 3: if $\beta_1 < 0$ and $\beta_2 = \beta_3 = 0$, a decreasing linear relationship exists between e and y.
- Scenario 4: if $\beta_1 > 0$, $\beta_2 < 0$, and $\beta_3 = 0$, there is an inverted U-shaped relationship that exists between e and y satisfying the EKC hypothesis as shown in Fig. 1.21. In this case the turning point income level can be estimated by $y_0 = -\beta_1/2\beta_2$ if the polynomial model of Eq. (1.2) is considered and by $y_0 = \exp(-\beta_1/2\beta_2)$ if the logarithmic model is used.
- Scenario 5: if $\beta_1 < 0$, $\beta_2 > 0$, and $\beta_3 = 0$, there is a U-shaped relationship that exists between e and y.
- Scenario 6: if $\beta_1 > 0$, $\beta_2 < 0$, and $\beta_3 > 0$, there is an N-shaped relationship that exists between e and y.
- Scenario 7: if $\beta_1 < 0$, $\beta_2 > 0$, and $\beta_3 > 0$, there is an inverted N-shaped relationship that exists between e and y.

The model of Eq. (1.2) is an empirical model since it requires historical data to determine the correlation coefficients and ultimately the relationship between the energy consumption and the economic output. The model, however, does not allow to identify the specific underlying structural drivers that lead to the relationship. Specifically, the inverse model of Eq. (1.2) cannot be utilized to predict future impacts of the implementation of energy policies in various sectors on the economic output. A deterministic analysis framework using the concept energy productivity is introduced in Chapter 5 to assess the effects of various energy efficiency programs and strategies on both future energy consumption and future economic output at the macroeconomic and microeconomic scales.

1.2 ENERGY EFFICIENCY POLICIES

A study was completed to rate the energy policies of over 111 countries worldwide. The criteria as energy policies and the administrative procedures considered in the analysis are summarized in Table 1.1. The study has found that most countries are making some efforts to improve the accessibility and the sustainability of energy (RISE, 2016). However, several energy policy shortcomings have been identified especially for lesser-developed countries:

- Access to energy and especially electricity in some countries remain low due mostly to the high cost of connecting to the grid. For instance, over 600 million people in the sub-Sahara countries remain without access to electricity. The cost for connecting to the electrical grid in several of these countries can reach $500 exceeding the worldwide average of $100. Poor families in some of these African countries live on an income of less than $2.00 per day.
- Several countries overlook energy efficiency and tend to pursue more aggressively renewable energy. However, implementing energy efficiency

TABLE 1.1 Criteria, Energy Policies, and Administrative Considered in the RISE Study (RISE, 2016)

Criteria	Energy policies and regulations	Administrative procedures
Energy accessibility	• Develop, implement, and monitor energy (electrification) plans • Establish a clear framework for grid electrification, minigrids, and stand-alone systems • Ensure access to affordable energy (electricity)	• Allow grid connection to customers including households • Permit new minigrids
Energy efficiency	• Establish energy efficiency plans • Create entities to overview and regulate energy efficiency plans • Develop and implement mandatory regulations and incentives for energy efficiency (i.e., minimum energy performance standards, building energy codes, energy labeling systems, and carbon pricing and monitoring) • Provide financial mechanisms for energy efficiency	• Establish enforcement mechanisms and procedures
Renewable energy	• Establish legal framework for renewable energy • Provide incentives and regulatory support for renewable energy • Allow grid connection and network access • Provide financial mechanisms for renewable energy	• Permit new renewable energy projects

programs remains globally more cost-effective than investing in renewable energy. The RISE study has noted that only a third of the countries worldwide have made progress in improving the energy efficiency of their economy by labeling energy-efficiency appliances or establishing building energy codes.

- Moreover, several countries are solely relying on grid extension to enhance electricity accessibility to their citizens and are missing the solar revolution. Indeed, development of low-cost PV systems offers today distributed energy resources and decentralized off-grid options that can accelerate the pace of electrification in many remote regions of the world.

1.2.1 Benefits for Improving Energy Efficiency of Buildings

According to the International Energy Agency (IEA, 2013), harnessing the full capacity for energy efficiency in the building sector could boost cumulative global economic output through 2035 by $18 trillion, with related benefits including the following:

- For building owners and occupants: improved durability, reduced maintenance, greater comfort, lower costs, higher property values, increased habitable space, increased productivity, and improved health and safety.
- For governments: improved air quality, reduced societal health costs, improved tax base, increase in GDP, and enhanced energy security. Moreover, energy efficiency can stretch the availability of nonrenewable energy resources, typically the main income source for several countries.
- Utilities: cost and operational benefits associated with decreased customer turnover, improved generation and transmission efficiency, and reduced system capacity constraints.

Moreover, improving energy efficiency of buildings can provide significant job creation potential often higher than that associated with investing in renewable energy. In particular, jobs created by the building energy efficiency sector require broader and often specialized skill sets than those needed for renewable energy. In particular, the job creation potential for improving the energy efficiency of the built environment including retrofitting existing buildings can provide significant social and economic benefits. A study by the Rockefeller Foundation (2012) indicated that retrofitting existing buildings including replacing and upgrading energy consuming systems offer an important investment opportunity with significant economic impacts including high employment potential. Specifically, the study has found that $279 billion can be invested in retrofitting US residential, commercial, and institutional buildings with savings in energy costs of $1 trillion over 10 years (i.e., with an average payback period of less than 4 years) resulting in an annual reduction by 30% in US electricity expenditures. When all the retrofits are undertaken, about 3.3 million jobs can be created throughout the US with the vast majority related to improving

TABLE 1.2 Employment Creation Impacts for Various Energy Sources by $1 Million of Expenditures

Energy source	Direct job creation per $1 million expenditures (number of job-years)	Indirect job creation per $1 million expenditures (number of job-years)	Direct and indirect job creation per $1 million expenditures (number of job-years)
Fossil fuel			
Oil and natural gas	0.8	2.9	3.7
Coal	1.9	3.0	4.9
Energy efficiency			
Building retrofits	7.0	4.9	11.9
Mass transit	11.0	4.9	15.9
Smart grid	4.3	4.6	8.9
Renewables			
Wind	4.6	4.9	9.5
Solar	5.4	4.4	9.8
Biomass	7.4	5.0	12.4

the energy efficiency of residential buildings. A job creation model has been developed by Pollin et al. (2009) based on survey data to estimate the number of direct and indirect full time jobs. Table 1.2 lists the number of job years that can be created from $1 million expenditures in the energy sector for various energy sources using the model of Pollin et al. (2009). For building retrofits, the direct effects consist of jobs created to implement the energy efficiency measures while the indirect effects are associated with the jobs needed to produce and supply energy efficiency equipment and materials. As noted by Pollin et al. (2009), there is an induced job creation associated with the direct and indirect effects, that is, the expansion of employment as the result of spending earnings from people who have acquired direct or indirect jobs. For the energy sector, the level of induced job-year creation is found to be 40% of the level for direct and indirect job-year creation. Thus, the induced job years created from building retrofits are estimated at 4.9 per $1 million of expenditures. Most of the jobs created in building retrofits are in the construction and manufacturing industries with a wide range of pay level and technical specialization including electricians, HVAC (heating, ventilating, and air conditioning) technicians, insulation installers, energy auditors, building inspectors, and construction managers. The total direct and indirect job creation due to $1 million expenditures is 11.9 job-years for the building retrofit while it is only 9.8 job-years for solar systems installation as indicated in Table 1.2.

1.2.2 Barriers for Energy Efficiency

However, securing the expected benefits from energy efficiency can be complex and challenging for several countries. Indeed, the building sector is highly fragmented, conservative, and slow to change. In particular, the building sector faces the following challenges in most countries:

- Multiple groups of stakeholders, with no unifying incentive to prioritize energy efficiency.
- Wide variations in technical education and energy-efficiency knowledge between individuals and groups.
- National and regional differences and inconsistencies in regulatory systems.
- Economic policies for new and retrofit building projects that often impede viable energy efficiency investments.

Several studies and pilot projects have shown that there are four core areas that typically limit the market penetration of energy efficiency:

- Lack of awareness and leadership particularly related to challenges in making the business case foe energy efficiency. Several proven and cost-effective energy efficiency systems are available for buildings. However, most decision-makers are not aware of these systems as well as of recent advances in building energy efficiency technologies.
- Workforce capacity and the need for proper skills and collaboration throughout the value chain to consult, plan, implement, and operate the right solutions. Indeed, several stakeholders are involved in designing, constructing, and operating buildings. It is difficult to achieve high energy efficient buildings without an integrated process between design, construction, and occupancy as well as collaboration effort between various stakeholders.
- Lack of adequate financing models that overcome the split incentives inherent in the building sector and enable value sharing. In particular, initial design decisions of new buildings are typically made by developers and not by owners or occupants who are responsible for paying the energy costs of using the buildings.
- Lack of consistent and long-term policy frameworks (national, subnational, regional, and city), including regulations and incentive schemes. For instance, several countries highly subsidize energy prices and often make investments in energy efficiency that is not cost-effective to building operators and owners. Table 1.3 summarizes the total subsidies for various energy resources reported for the world during 2013–2015. Table 1.4 lists the highest energy subsidy per capita level as well as the subsidization rate reported for countries during 2015.

TABLE 1.3 World Global Subsidies for Energy Resources for 2013–2015 Expressed in Billions of 2015 Real US Dollars (IEA, 2017)

Energy resources	2013	2014	2015
Oil	263,504	242,258	144,161
Electricity	124,578	120,438	100,228
Natural gas	103,020	95,201	76,280
Coal	2016	1622	1181
Total	493,117	459,520	321,850

TABLE 1.4 Top Energy Subsidies Levels by Countries During 2015 (IEA, 2017)

Country	Average subsidization rate (%)	Subsidy per capita ($/person)	Total subsidy as share of GDP (%)
Kuwait	70%	1547.4	5.0%
Saudi Arabia	72%	1542.3	7.4%
Qatar	52%	1508.0	1.8%
Bahrain	52%	1211.9	5.5%
UAE	40%	1082.4	2.9%
Turkmenistan	70%	1021.8	15.4%
Libya	65%	668.0	11.0%
Iran	71%	662.4	13.5%
Venezuela	91%	640.6	8.3%
Trinidad and Tobago	33%	497.4	2.8%
Brunei	30%	422.1	1.5%
Algeria	74%	371.9	8.2%
Argentina	35%	241.1	1.8%
Russia	22%	210.9	2.3%
Uzbekistan	57%	206.8	9.8%
Kazakhstan	25%	180.3	1.8%
Iraq	45%	164.7	3.5%
Egypt	48%	147.0	4.1%
Ecuador	27%	139.0	2.3%
Ukraine	31%	137.7	6.9%

GDP, Gross domestic product.

In the following sections, mechanisms and regulatory strategies to improve energy efficiency of new and existing buildings are described.

1.2.3 Energy Efficiency Programs for New Buildings

The main strategy commonly employed by policy makers to improve the energy efficiency of new buildings is a set of standards and codes. In particular, minimum energy performance standards (MEPS) are often utilized as strategies to ensure only high-efficiency building products and equipment are manufactured, sold, and installed. MEPS can be specific to appliances (i.e., refrigerators, freezers, dishwashers, washing machines, dryers, and ovens), lighting fixtures, and heating and cooling systems. Table 1.5 illustrates an

TABLE 1.5 **Labels for Minimum Energy Performance Standards for Refrigerators, Freezers and Air Conditioners, and Washing Machines for KSA (Krarti et al., 2017)**

Star rating	Air conditioners (EER = 3.412 COP expressed in Btu/Wh)	Refrigerators/freezers (percent of energy consumption relative to a baseline)	Washing machines (function of energy use per load capacity)
1	<7.5	5%	<2.0
2	7.5–8.5	10%	2.0–2.9
3	8.5–9.0	15%	3.0–3.9
4	9.0–9.5	20%	4.0–4.9
5	9.5–10.0	25%	5.0–5.9
6	10.0–11.5	30%	>6.0
7	11.5–12.4		
7.5	12.4–13.4		
8.0	13.4–14.5		
8.5	14.5–15.6		
9.0	15.6–16.8		
9.5	16.8–18.1		
10	<18.1		

Note: SASO has updated the energy performance standards for washing machines (conforming to star rating 4 and above only), refrigerators (conforming to star rating 1 and above only), and air-conditioners (conforming to star rating 3 and above can only be sold and manufactured).
Source: SASO, 2012. Energy Labelling and Minimum Energy Performance Requirements for Air-Conditioners (Standard No. 3459). Saudi Arabia Standards Organization, Saudi Arabia. SASO, 2013. Energy Labelling Requirements of Household Electrical Clothes Washing Machines (Standard No. 3569). Saudi Arabia Standards Organization, Saudi Arabia. SASO, 2014. Energy Performance Capacity and Labelling of Household Refrigerators, Refrigerator-Freezers, and Freezers (Standard No. 3620). Saudi Arabia Standards Organization, Saudi Arabia.

example of MEPS for appliances set recently in KSA (Krarti et al., 2017). Moreover, the establishment of voluntary or mandatory building energy efficiency codes (BEECs) has been proven to be an effective strategy in several countries to compel building industry stakeholders to design and construct an energy efficient and sustainable built environment. Incentive programs have been considered to encourage building owners and developers to adopt even stricter regulations than applicable MEPS and BEECs to improve the energy performance of new buildings. These incentives can be monetary and nonmonetary as indicated in the following options considered by select countries:

- Full or partial grants (US, Singapore) to cover any additional design costs, or costs beyond meeting BEEC requirements, as well as for demonstration projects involving untested and novel technologies.
- Subsidies for loans or interest rates (Germany, US, Austria, Japan).
- Energy efficiency or green mortgages (US, Mexico).
- Tax benefits (US) using several options such as (1) reductions in import tax duties for energy-efficiency equipment, (2) income tax deductions, and (3) credits for energy efficient appliances and equipment.
- Nonmonetary incentives (US, Korea) through expedited construction permits and/or relaxed zoning restrictions (size, density).
- Awards (US) using Energy Star labels.
- Rating systems LEED (US) and BREAM (UK).

The BEEC establishment process for MEPS and BEECs can be lengthy and involved with development, implementation, enforcement, and revision phases. The following sections outline the type of BEECs and their establishment procedures.

1.2.4 Types of Building Energy Codes

BEECs are also known as energy standards for buildings, thermal building regulations, energy conservation building codes, or simply energy building codes. Energy policy makers and governments utilize BEECs as regulatory mechanisms to reduce energy consumption and environmental impacts of buildings while maintaining acceptable indoor environmental quality. Typically, BEECs consist of mandatory design requirements to improve the energy performance of buildings. Since the 1970s, BEECs have been developed and implemented in several countries. For some countries, it has been shown that BEECs are effective in reducing the energy consumption for buildings. In particular, the implementation of mandatory BEECs has resulted in energy use reduction of households for most EU countries. Indeed, the reduction in energy consumption for residential buildings has ranged from 22% in Germany and the Netherlands to 6% in southern European countries (IEA, 2013).

Two approaches are commonly considered to develop BEECs specific to new buildings:

- *Prescriptive-based approach*: In this case, the BEECs include sets of minimum energy performance requirements for each component of the building: windows, walls, and heating and cooling equipment. Two compliance paths can be considered: (1) each building component has to meet strict minimum energy performance requirements such as minimum thermal performance levels for walls and windows (U-values and SHGC-values), and (2) trade-offs between the energy performance requirements of different components. While the prescriptive-based approaches are simple to implement and enforce, they lack flexibility and do not encourage an integrated design of buildings. Table 1.6 illustrates examples of current BEECs following the prescriptive approach for selected MENA countries (RCREEE, 2015).
- *Performance-based approach*: BEECs based on performance set requirements for the overall building energy consumption. Thus, performance-based BEECs encourage an integrated design approach to account for interactions between building components to optimize the energy performance for the entire building. Several options for the performance-based BEECs can be developed and implemented including the following:
    - Partial performance of subsystems (building envelope using the overall thermal transfer value or OTTV analysis approach)
    - Performance of multiple subsystems (maximum energy use of demand for lighting and air conditioning systems)
    - Total building performance (energy consumption or energy cost) using three options:
        - Fixed budget approach using a fixed energy consumption such as kWh/m^2 (key assumptions are required including operation settings and internal loads).

TABLE 1.6 **Prescriptive Requirements for Building Envelope for Selected MENA Countries**

Country	Minimal *R*-values Roofs (m^2°C/W)	Minimal *R*-values Walls (m^2°C/W)	Minimal *R*-values Floors (m^2°C/W)
Egypt	1.67	1.0	–
Kingdom of Saudi Arabia	1.75	1.0	–
Lebanon	1.75–2.27	0.48–1.85	0.40–1.82
Tunisia	1.33–1.82	0.83–1.67	–
Jordan	0.51–1.75	0.56	–
Kuwait	2.50	1.75	

MENA, Middle East and North Africa.

TABLE 1.7 Performance Requirements for Office Buildings in Tunisia

Building thermal performance class	*BECTh* value (kWh/m^2 yr)
Class 1	$BECTh \leq 75$
Class 2	$75 < BECTh \leq 85$
Class 3	$85 < BECTh \leq 95$
Class 4	$95 < BECTh \leq 105$
Class 5	$105 < BECTh \leq 125$
Class 6	$125 < BECTh \leq 150$
Class 7	$150 < BECTh \leq 180$
Class 8	$BECTh > 180$

- Custom budget approach using comparative analysis with a reference building (typically, the reference building is the same as the proposed building but complies with a set of prescriptive requirements).
- Points system using a score system for different energy efficiency measures.

Table 1.7 illustrates the performance-based BEEC requirements for office buildings in Tunisia (Ihm and Krarti, 2012). As indicated in Table 1.7, the BEEC requirements are set-based ratings or classes using the annual energy needs allowed for heating and cooling loads (expressed in kWh/m^2 yr). The building has to meet a specific energy rating depending on the climate zone. The energy rating-based BEEC permits an easy revision process of the energy efficiency requirements by simply changing the class levels.

1.2.5 Overview of BEEC Establishment

To be effective in improving the energy efficiency of buildings, BEECs have to go through several phases including development, implementation, enforcement, and revision as summarized in Table 1.8. Indeed, the establishment cycle of BEECs may take over 6 years to be adequately implemented and enforced.

The development process for BEECs varies widely between countries. However, the following steps are typically considered to develop an effective BEEC:

Step 1: Perform a review and an analysis of existing international BEECs.
Step 2: Assemble a working group made up of experts to assist in the BEEC development.
Step 3: Survey representative samples of national building stock through walk-through energy audits and questionnaires.
Step 4: Develop prototypical building energy models as benchmarks for BEEC development.

TABLE 1.8 Typical Phases and Their Duration for Establishing Building Energy Codes

Phases for building energy efficiency code establishment	Year								
	1	2	3	4	5	6	7	8	9
Code development									
Code implementation									
Code enforcement									
Code revision									

Step 5: Collect climatic data, available energy efficiency technologies, and costs of adopting these technologies.
Step 6: Perform technical and economic analyses using state-of-the-art computer simulation tools to assess best energy efficiency measures based on energy savings and cost-effectiveness. The most effective measures are typically included as part of the BEEC requirements.
Step 7: Draft, revise, and finalize BEEC requirements based on public review to include key stakeholders.
Step 8: Adopt BEEC officially as mandatory or voluntary code (selecting if BEEC should be mandatory or voluntary depends often on existing enforcement mechanisms of general building codes including structural, electrical, mechanical, and fire codes).

While designing and developing BEECs, the following features could be considered to maximize their effectiveness:

- *Simplicity and/or flexibility:* BEEC can include both prescriptive and performance approaches (a prescriptive approach is preferred as an initial step for BEEC adoption especially for small buildings).
- *Accuracy:* BEEC has to be technically accurate and consistent (prescriptive requirements should be consistent with performance compliance requirements).
- *Affordability:* BEEC should account for the availability of energy efficiency technologies (the costs of these technologies should be sufficiently low to make them cost-effective).
- *Beneficial impact:* BEEC should provide tangible benefits to owners, occupants, and society (life-cycle cost or LCC analysis should be used as part of the BEEC development).

The implementation approach of a BEEC is often an important step and requires adequate preparation. The following implementation approach is suggested:

- Establish BEEC administration and enforcement structure, typically through a government agency.

- Define a clear BEEC compliance process through easy-to-use forms or computer tools.
- Develop training and building capacity programs for code officials, designers, architects, engineers, contractors, and suppliers.
- Establish outreach programs to the building industry and the public.
- Initiate, complete, and evaluate demonstration programs through incremental financial incentives in the initial phase of BEEC implementation.
- Set firm periods and dates for implementation of the BEEC requirements with clear procedures and mechanisms.
- Identify institutions and mechanisms to evaluate BEEC impact and effectiveness in reducing energy consumption.

The enforcement phase of BEECs is often the most difficult step to complete successfully especially for countries with limited qualified institutions and established regulations and standards related to constructing new buildings. Some of the enforcement options and their associated implementation requirements utilized by some countries are outlined below:

- Government agencies (US): This option requires well-trained inspectors and may result in high operating costs for the government.
- Private agencies (France, Mexico): This option allows a third party, certified by the governmental institutions, to enforce BEEC requirements. A training program for private enforcement agencies needs to be established. The costs incurred by the government to implement this option are generally moderate.
- Self-certification to owners (Germany): This option relies on the builders themselves to ensure compliance with the BEEC requirements typically through "verification of compliance" statements and forms. The costs for implementing this enforcement option are generally low and are mainly associated to the training and certification programs.

1.2.6 Beyond BEEC Requirements

Several countries have set regulations and programs to improve building energy efficiency and design low-energy and sustainable buildings beyond the requirements of BEECs. As will be discussed throughout this book, current technologies allow to design and retrofit cost-effectively buildings to be near zero-energy. The main strategy to push designing and retrofitting high-energy performance buildings beyond BEECs involves rating and labeling systems. Some of the sustainable building rating systems include the following:

- LEED: Leadership in Energy and Environmental Design, is the first rating system for buildings and was developed in 1994 by architects and engineers.
- UK BREEAM: Building Research Establishment's Environmental Assessment Method, is used mostly in the UK and was developed in 1990.

- CASBEE: Comprehensive Assessment System for Building Environmental Efficiency, was developed in Japan in 2001.
- GBTool: Green Building Tool, an international (25 countries) rating system was developed in 1998.
- Green Globes: was developed for Canada using BREEAM and Green Leaf in 2004. It is intended to be an on-line tool.
- Green Mark Rating: Building and Construction Authority (BCA) launched this rating system in 2005 in Singapore for new and existing buildings.

1.2.7 Holistic Design Approach

Fig. 1.25 shows the conventional structure defining the interactions between various members of a design team for a new building. While the architect typically plays the role of a coordinator for the design team, several professionals and engineers have crucial involvement in various phases of the building design and construction. The role and the main duties of each team member as listed in Fig. 1.10 based on the conventional design process are briefly described as follows:

- The architect generally develops the overall aesthetic form and structure of the building including its shell and material requirements while accounting for factors, concepts, and practices that ensure functional, safe, and economical design as per the client specifications. For most current building projects, the architect delivers the design through plans and drawings completed using computer-aided drafting and building information modeling tools. Specifically, the architect coordinates the project and communicates the design specifications to various engineers through computer-aided drafting and/or building information modeling drawings. After the design phase,

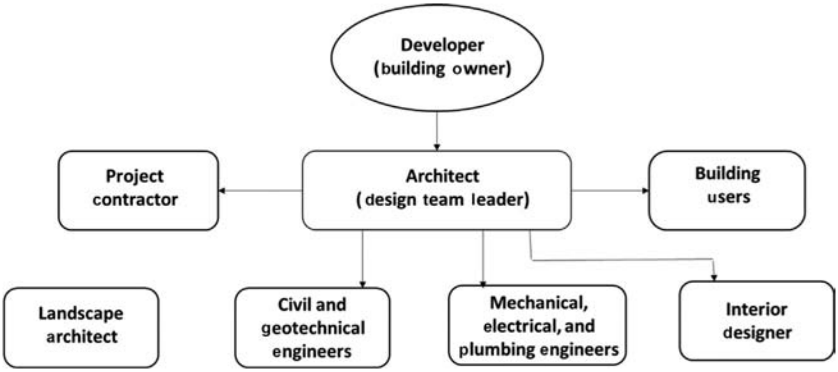

FIGURE 1.25 Interactions between team members of a conventional design process for new buildings.

the architect continues overseeing the project during the construction phase through site visits and some revisions of the design based on any changes to account for the client needs, budget constraints, and other unexpected factors. Through consultations with the architect, engineers provide detailed design specifications for various systems within the building from the conception to the construction phases.

- The mechanical engineer develops design solutions and specifications for energy systems including HVAC, plumbing, and fire protection. Moreover, the mechanical engineer assists the architect in specifying the thermal properties of materials to be used in the building structure as well as any energy systems to improve the energy performance of the building to meet any relevant codes and standards and/or labeling systems.
- The electrical engineer provides design options for power service and distribution as well as lighting, communication, fire detection and alarm, as well as general electrical equipment and associated space requirements. Moreover, the electrical engineer collaborates closely with the architect and the mechanical engineer to meet requirements of any applicable codes and standards and any energy performance targets such as integrating cost-effective energy efficiency technologies including cogeneration and distributed generation systems such as roof-mounted photovoltaic panels.
- The geotechnical or civil engineer evaluates and provides recommendations for the geotechnical specifications of the building site such as ground soil properties and design solutions for earthworks (i.e., grading, drainage, and pavement). In addition, the civil engineer assists in the design of building foundations to ensure that these building elements meet applicable codes and standards. Similarly, the structural engineer assists the architect in the planning, design, and construction of the building structure and its components such as slabs, beams, columns, and foundations. Moreover, the structural engineer specifies the various materials for the building shell following relevant codes and standards.
- The interior designer and the landscape designer work mostly with the client and the architect. The main role of the interior designer is recommending layouts and types of furnishings and decorations within the interior building spaces. The main task of the landscape designer is to propose layouts and types of greenery to be located just outside the building structure. Both designers may also have to communicate their plans and service needs to the mechanical and electrical engineers.

The conventional design process outlined in Fig. 1.25 has several significant shortcomings that often impede designing high performance buildings. In particular, the process limits the involvement of all the stakeholders in the early conception and design phases of the project. For instance, the mechanical engineer is requested to size the mechanical system only after major architectural and structural decisions have been made. Moreover, and due to

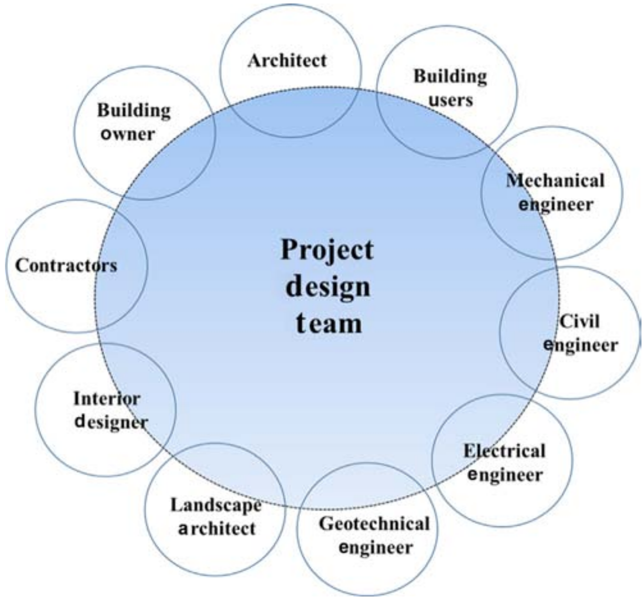

FIGURE 1.26 Interactions between building stakeholders in an integrated design process for new buildings.

limited time and budget given to some systems (i.e., lighting and electrical), innovative and boilerplate design specifications are adopted. Some of these shortcomings can be avoided through an integrated design process as illustrated in Fig. 1.26.

Indeed, the integrated design process allows multiple disciplines and most of the building stakeholders to be engaged and constructively participate in defining and selecting the main specifications of the building. The engagement is facilitated through a series of meetings and discussions. These meetings, also referred to as “charrettes,” are especially important in the early design phase to ensure that all relevant building stakeholders including occupants and contractors are able to discuss and share ideas and alternatives. The integrated design process has proven to be effective in achieving higher energy performance and lower costs for sustainable facilities such as LEED-rated or Energy Star-labeled buildings. As will be discussed throughout this book, a wide range of strategies and systems can be utilized to design and retrofit buildings to be more energy efficient. Indeed, several factors affect the overall performance of buildings including mechanical and electrical systems, building occupants, sustainability targets, climate conditions, as well as construction and operation costs. With an integrated design approach, these factors can be considered by the integrated design team members to determine the best synergies and alternatives to achieve the highest levels of energy performance for the new buildings.

1.3 STRATEGIES FOR LARGE SCALE EE PROGRAM IMPLEMENTATION

Three strategies are commonly adopted to develop and implement successfully large-scale energy efficiency programs for existing building stocks:

1. Foundational strategies, also called no-regrets strategies: activities needed to ensure large penetration of the energy efficiency programs.
2. Voluntary strategies: recommended activities to support upgrade projects for all categories of the building stock.
3. Mandatory strategies: needed activities to achieve large-scale upgrade of the existing building stock.

The basic tasks and objectives for each of the listed strategies are described in the following sections.

1.3.1 Foundational Strategies for Large-Scale Energy Efficiency Program

Some of the main phases for foundational strategies to develop large-scale energy efficiency programs for the existing building stock include the following:

- *Phase 1*: Data collection and analysis of energy performance and characteristics of the building stock. In this phase, energy consumption databases for existing buildings should be developed and maintained. The databases can be utilized to establish baselines and benchmarks for the energy performance of existing building stocks. These baselines are then used to develop performance metrics and to document the impacts of any future energy retrofit projects. The primary stakeholders to implement this phase are government agencies, utilities, nonprofit organizations, and building industry professionals. For instance, energy use intensity (EUI) values such as those outlined in Table 1.9 for various building types in France and the US are useful to assess the energy performance of existing buildings within a country or a state (Krarti, 2011).
- *Phase 2*: Support for standards development, compliance, and enforcement. In this phase, programs and standards for improving the energy performance of existing buildings are developed, implemented, and monitored. In particular, administrative structures should be established and the required financial support should be provided for the development and implementation of the programs. Moreover, clear monitoring compliance procedures should be developed to ensure successful enforcement of the implemented energy efficiency standards and programs. Depending on the country, the main tasks for this phase can be completed using government agencies, utilities, building industry professionals, as well as equipment manufacturers and distributors.

TABLE 1.9 **Energy Use Intensity (EUI) Values Expressed in kWh/m^2 for Various Building Types in France and the US**

Major building activity	France	US
Office	296	300
Education	160	270
Health care	283	610
Lodging	230	325
Food service	416	649
Mercantile and service	264	250
Public assembly	NA	305
Warehouse and storage	NA	147

- *Phase 3*: Education and outreach to building owners and managers. To successfully implement the adopted energy efficiency programs, it is important to educate building owners and managers about the desired goals and the set requirements. In particular, a market research should be carried out to understand customers' attitudes toward energy efficiency specific to existing buildings. Generally, energy-intensive buildings should be first targeted to maximize the benefits of the programs. In addition, a series of informational sessions about the benefits of energy efficiency should be organized to reach building owners, real estate agents, and facility managers. Some of the stakeholders that can assist in the education and outreach efforts include government agencies, utilities, education partners, building industry professionals, and realtor associations.
- *Phase 4*: Training of skilled workforce. This phase is important to ensure the success of any energy efficiency program for existing buildings. Indeed, a skilled workforce is needed to execute any level of building energy retrofits. Due to the wide range of skills required, specific technical training programs have to be developed and delivered possibly through existing educational institutions. These programs should train workforce and contractors in specific energy efficiency technologies suitable for buildings (i.e., high efficacy lighting fixtures, high performance heating and cooling systems, and intelligent controls). These programs should be scheduled regularly to allow practitioners and professionals to update their technical skills. The main stakeholders that can develop and deliver technical and practical training programs include government agencies, utilities, education institutions, and building industry professionals.

Once the foundational strategies for large-scale energy efficiency programs have been developed and tested, two paths for implementation can be considered either voluntary or mandatory. It is recommended to start first with voluntary strategies followed by mandatory strategies.

1.3.2 Voluntary Strategies for Large-Scale Energy Efficiency Program

The development and implementation of voluntary strategies to improve the energy performance of existing buildings include four phases.

- *Phase 1*: Develop a wide range of pathways to achieve energy efficiency. In particular, building owners and managers should have a wide range of options to implement energy efficiency upgrades. For instance, retrofit options could include single energy efficiency measures (i.e., to replace a heating or cooling system), multiple measures (i.e., to upgrade the insulation level in the walls as well as to replace the windows), and deep-retrofits (i.e., to ensure the building meets any current energy-efficiency codes and standards). The contractors need to be informed and educated about available energy efficiency program offerings for the existing building stock as well as about procedures used to document performance metrics of retrofit projects. The main stakeholders for this phase include constructors, building owners, facility managers, realtor associations, and financial institutions.
- *Phase 2*: Develop standardized analysis tools for benchmarking, auditing, and retro-commissioning. These analysis tools are important to facilitate the implementation and monitoring of the energy efficiency programs. Various tools, analysis methods, and protocols are needed depending on the desired objectives including the following:
  - Benchmarking tools to establish baseline metrics for different categories of the existing building stock (offices, hotels, restaurants, retails).
  - Reliable tools and methods for calculating energy use and cost savings for a wide range of building categories and energy efficiency measures.
  - Easy-to-use and cost-effective energy audit protocols and tools.

  The main stakeholders for developing these tools include government agencies, research and educational institutions, building industry professionals, and software developers.
- *Phase 3*: Develop innovative financing mechanisms of energy efficiency upgrades. Building owners and managers often need to have access to financing mechanisms to improve the overall energy performance of their buildings. Several options should be provided to obtain the required capitals through (1) engaging financial institutions to provide loans for energy efficiency projects, (2) encouraging third party private agencies to invest in energy efficiency projects such as energy service companies, and (3) establishing monetary incentives for energy efficiency retrofit projects (such as energy efficiency mortgages and tax reductions). Government agencies,

financial institutions, energy service companies, and nonprofit organizations are typically the main stakeholders for this phase.

1.3.3 Mandatory Strategies for Large-Scale Energy Efficiency Program

To make energy efficiency upgrades for existing buildings mandatory, the following steps can be considered:

- First require minimum energy efficiency levels for public buildings. Governments should provide an example to the private sector by improving the energy efficiency of their buildings.
- Require minimum efficiency levels for rental properties. It is important to ensure that owners provide healthy and energy efficient buildings for renters.
- Require compliance with minimum building energy efficiency for deep retrofits. Typically, deep retrofitted buildings can be required to meet the same energy standards and regulations set for new buildings.

To assist in implementing mandatory programs for existing buildings, several stakeholders can be involved including government agencies, utilities, building industry professionals, property owners, nonprofit organizations, inspectors, building owners, and facility managers.

1.3.4 Approaches for Energy Intensive Buildings

A skilled workforce with a wide range of technical and expertise areas is needed to properly improve the energy performance of existing buildings. Some of the desired expertise areas include the following (Krarti, 2011):

Energy Performance Contracting (EPC): The EPC mechanism has been proven to be effective in financing energy efficiency projects for buildings. Typically, a third party such as energy service companies can evaluate, finance, and implement energy retrofit projects through agreements with building owners or operators. Often these agreements call for proven and sound approaches to measure and verify energy savings incurred from building retrofits. The third party (i.e., energy service companies) would receive payments based on measured and verified energy cost savings achieved by the building retrofit as illustrated in Fig. 1.27.

Demand-Side Management (DSM): This type of management program is typically considered by electric utilities as a mechanism to reduce their customers' peak demands. Moreover, DSM programs are employed by electric utilities to prevent operation or construction of peak generation or spot-market purchases that are often not efficient and are costly. To entice their customers, utilities provide monetary incentives for delivered demand and energy savings.

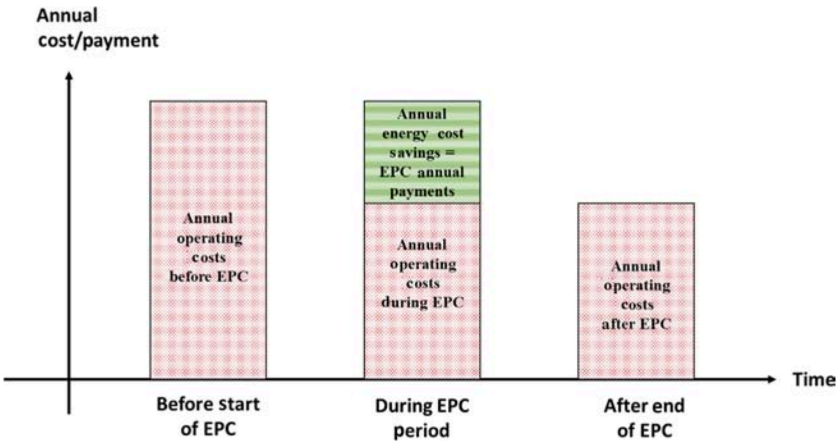

FIGURE 1.27 Mechanism for energy performance contracting to finance building retrofit projects.

DSM programs can be targeted for residential and commercial buildings as well as industrial facilities and are typically operated by governmental agencies. There are several DSM programs including the following:

- retrofits (HVAC, lighting, motors, controls)
- new construction (integrated design assistance)
- retro-commissioning (no/low-cost operational)
- load curtailment (on call demand reduction for 3–4 h blocks on peak summer days)
- air-conditioner rebates for efficiency upgrades
- new construction (Energy Star homes)

Commissioning and Retro-Commissioning: For new buildings, commissioning programs can be used to assure that the project concept, design, and installation are appropriate and correct, and that the systems operate properly to the design specifications. For existing buildings, retro-commissioning programs are used to evaluate, troubleshoot, and optimize mechanical systems and their controls to increase system reliability, energy efficiency, occupant comfort, and indoor air quality. Moreover, recommissioning programs can be considered for existing buildings that have already been commissioned or retro-commissioned. In the last decade, continuous commissioning programs have become popular to ensure that there is always an ongoing process to resolve operating problems, minimize comfort complaints, and optimize energy performance for existing buildings. Table 1.10 provides an example to illustrate the difference between commissioning and capital investment projects.

TABLE 1.10 Example to Illustrate the Difference Between Commissioning and Capital Improvement Programs

Option	Capital improvement	Retro-commissioning
Retrofit measure	Replace existing chillers with high-efficiency equipment with rated efficiency from 0.59 kW/ton to 0.43 kW/ton	Identify and implement low-cost improvements for operating the chillers such as fix broken chilled water valves
Energy use savings	160,000 kWh/yr	530,000 kWh/yr
Cost savings	$5700/yr	$19,200/yr
Project cost	$240,000	$3800
Simple payback period	42 years	0.2 year

Measurement and Verification (M&V) Analysis: M&V protocols are required to assess and estimate the energy savings associated to any energy retrofit project especially when energy performance contracts are involved. Several techniques are available to determine the energy savings through the use of inverse or forward models based on short-term or long-term metered data or simple utility data. In particular, the International Performance Measurement and Verification Protocol provides four options for M&V analysis depending on the retrofit project type. In this section, these four options are briefly described including their methods of implementation and common applications (Krarti, 2011):

- Option A: Suitable for energy efficiency measures applied to isolated systems or equipment.
  - Method: Spot measurements, short-term monitoring, simple calculations.
  - Applications: Single energy efficiency measures with known impacts (i.e., lighting fixtures, motors with constant speed).
- Option B: Suitable for single energy efficiency measures.
  - Method: Similar to Option A but only end-use measurements are performed and/or specific values of parameters are identified.
  - Applications: Installations of complex systems (i.e., chillers, boilers, motors with variable speed drives).
- Option C: Suitable for energy retrofit applied to the entire building.
  - Method: Analysis of utility data and/or monitored energy use of entire building.
  - Applications: Acceptable for energy efficiency projects with large expected energy use savings (i.e., weatherization measures including adding thermal insulation to the building envelope). Typical savings of 20% in total building energy consumption.

- Option D: Suitable when detailed simulation and calibrated models are required to estimate energy savings.
  - Method: Modeling of buildings using detailed simulation tools such as eQUEST, EnergyPlus, and others. The energy models have to be calibrated using available metered and/or utility data.
  - Applications: Projects with several energy efficiency measures such as the case for deep retrofits.

Energy Auditing: Energy audit is an important procedure for improving the energy efficiency of existing buildings. Energy auditors require multidisciplinary skills and technical knowledge about all building systems including envelope, lighting, electrical, mechanical, and controls. Moreover, energy auditors should be able to perform a detailed economic analysis to assess the cost-effectiveness of various energy efficiency measures as well as the impact of utility rate selection on annual energy costs. There are four types of energy audits as outlined in the following sections (Krarti, 2011).

Utility Data Analysis: The main purpose of this type of audit is to carefully analyze the operating costs of the facility. Typically, the utility data over several years are evaluated to identify the patterns of energy use, peak demand, weather effects, and the potential for energy savings. In particular, it is important that the energy auditor understands clearly the utility rate structure that are or could be applied to the facility for several reasons including the following:

- Checking the utility charges and ensuring that no mistakes were made in calculating the monthly bills. Indeed, the utility rate structures for commercial and industrial facilities can be quite complex, with ratchet charges and power factor penalties.
- Determining the most dominant charges in the utility bills. For instance, peak demand charges can be a significant portion of the utility bill, especially when ratchet rates are applied. Peak shaving measures can then be recommended to reduce these demand charges.
- Identifying if the facility can benefit from using other utility rate structures or purchase cheaper fuel and reduce its operating costs. This analysis can provide a significant reduction in the utility bills, especially when considering time of use and real-time pricing rate structures.

Moreover, the energy auditor can determine if a facility is prime for energy retrofit projects by analyzing the utility data. Indeed, the energy consumption for buildings can be normalized and compared to known indicators such as EUI commonly used for commercial buildings.

Walk-Through or Level 1 Energy Audit: This audit consists of a short on-site visit to the facility to identify areas where simple and inexpensive actions can provide immediate energy and/or operating cost savings. Some engineers refer to these types of actions as operating and maintenance (O&M) measures. Examples of O&M measures include lowering heating set-point temperatures, replacing broken windows, insulating exposed hot water or steam pipes, and adjusting boiler fuel–air ratio.

Standard or Level 2 Energy Audit: The standard audit provides a comprehensive analysis of the energy systems of the facility. In addition to the activities described for the walk-through audit and the utility cost, the standard energy audit includes the development of a baseline for the energy use of the facility and the evaluation of energy use savings and cost effectiveness of appropriately selected energy efficiency measures. Generally, simplified tools are utilized in the standard energy audit to develop baseline energy models and to predict the energy savings of energy conservation measures. Among these tools are the degree-day methods and linear regression models. In addition, a simple payback period is often utilized as a metric to determine the cost-effectiveness of energy efficiency measures.

Detailed Level 3 Energy Audit: This is the most comprehensive but also time-consuming energy audit type. Specifically, the detailed energy audit includes the use of instruments to measure energy use for the whole building and/or for some energy systems within the building (for instance, by end uses including lighting fixtures, office equipment, appliances, fans, chillers, and boilers). In addition, sophisticated computer simulation programs are typically employed for detailed energy audits to evaluate and recommend energy retrofits for the facility. The computer simulation programs used in the detailed energy audit typically provide the energy use distribution by load type (i.e., energy use for lighting, fans, chillers, boilers, etc.). They are often based on the dynamic thermal performance of the building energy systems and usually require a high level of engineering expertise and training. Techniques available to perform measurements for an energy audit are diverse (Krarti, 2011). However, a rigorous economical evaluation of the energy efficiency measures is generally performed as part of a detailed energy audit. Specifically, the cost-effectiveness of energy retrofits may be determined based on the LCC analysis rather than the simple payback period analysis. As presented in Chapter 5, LCC analysis takes into account a number of economic parameters such interest, inflation, and tax rates.

Table 1.11 summarizes performance data and energy efficiency measures that are generally considered when conducting an energy audit of thermal and electrical systems for existing residential and commercial buildings.

1.4 SUMMARY

This chapter provides an overview of the current and future trends in the energy consumption in the world as well as in various countries and regions. The energy consumption specifically attributed to the building sector is described based on historical data as well as future projections. In addition, the chapter summarizes various energy efficiency strategies and policies suitable for improving the energy performance for new and existing buildings. General technical practice areas required to implement energy efficiency programs are outlined to pinpoint

TABLE 1.11 Performance Data and Energy Efficiency Measures for Existing Residential and Commercial Buildings

Thermal systems	Electric systems
Utility analysis	
Thermal energy use profile (building signature) Thermal energy use per unit area (or per student for schools or per bed for hospitals) Thermal energy use distribution (heating, DHW, process, etc.) Fuel types used Weather effect on thermal energy use Utility rate structure	Electrical energy use profile (building signature) Electrical energy use per unit area (or per student for schools or per bed for hospitals) Electrical energy use distribution (cooling, lighting, equipment, fans, etc.) Weather effect on electrical energy use Utility rate structure (energy charges, demand charges, power factor penalty, etc.)
On-site survey	
Construction materials (thermal resistance type and thickness) HVAC system type DHW system Hot water/steam use for heating Hot water/steam for cooling Hot water/steam for DHW Hot water/steam for specific applications (hospitals, swimming pools, etc.)	HVAC system type Lighting type and density Equipment type and density Energy use for heating Energy use for cooling Energy use for lighting Energy use for equipment Energy use for air handling Energy use for water distribution
Energy use baseline	
Review architectural, mechanical, and control drawings Develop a base-case model (using any baselining method ranging from very simple to more detailed tools) Calibrate the base-case model (using utility data or metered data)	Review architectural, mechanical, electrical, and control drawings Develop a base-case model (using any baselining method ranging from very simple to more detailed tools) Calibrate the base-case model (using utility data or metered data)
Energy conservation measures	
Heat recovery system (heat exchangers) Efficient heating system (boilers) Temperature setback EMCS HVAC system retrofit DHW use reduction Cogeneration	Energy efficient lighting Energy efficient equipment (computers) Energy efficient motors HVAC system retrofit EMCS Temperature setup Energy efficient cooling system (chiller) Peak demand shaving Thermal energy storage system Cogeneration Power factor improvement Reduction of harmonics

DHW, Domestic hot water; EMCS, energy management control system; HVAC, heating, ventilating, and air conditioning.

to the technical skills required to implement energy efficiency programs for the building sector. Moreover, the concept of energy productivity is introduced using economic output and energy consumption data for the world and select countries.

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